

Notes on the focus group: Socio-economic forces on a major architecture

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Background

This focus group was a follow up to the EuroPLoP 2005 focus group on *Conway's Law*. The 2005 group recognised that software architectures are created by many different forces. While software engineers focus on technical issues a lot of social and economic forces work to shape the architecture every bit as much as the technical issues.

The aim of this focus group was to expose and understand some of those forces.

Format

The focus group was once again organised and lead by Allan Kelly and Lise Hvatum. After an initial introduction from Lise describing the back ground and objectives of the group each participant was invited to make an opening statement describing their own thoughts and objective in attending the group.

With the preliminaries over the group adopted a structured conversation approach. Participants raised ideas that were discussed, these in turn lead to more ideas and discussion. We appropriate the organizers recorded suggestions on flip charts.

The notes which follow are taken from those flip charts and were appropriate, and possible, additional description is added.

Findings

Roles: *Who impact your architecture?*

- Architects who don't implement
- Informal decision makers, e.g. strong individuals in the wider organization
- Team member, e.g. product manager and developers

On your own project

1. Define and understand the social-economic forces
2. Look at (analyze) the forces
 - Balance the forces
 - Which forces are the most important?
 - Which forces are going to conflict?

- Which forces can't be change?
- Which forces are beyond your influence?
- 3. Communicate the socio-economic forces to others - who you communicate the forces to may vary.

Forces that impact the architecture

- Legal forces: cross cut the project
- Merging of companies and the merging of products
- Investment commitment, i.e. money
- Need for flexible business processes
- Technology hype - can define architecture at a deep level, e.g. SOA
- Strategic decisions
- Dependencies on other systems
- Dependencies on legacy systems
- Micro-architecting *what do we mean here?*
- IT people are not seen to make cost/business decisions in a conscious manor
- Self-fulfilling project scope, i.e. if you expect to build a big system you will end up building a big system and vice-versa
 - *This discussion returned the group to Conway's Law and the previous years focus group. (See EuroPloP 2006 proceedings.)*
- Budgeting
- Organizational flexibility: businesses may change from a stable strategy in order to pursue a business opportunity and will expect the systems to follow them
- Products that create change: can be good or bad
- Introduction of products to secure organizational entities (create whole)
- Flexibility provides by SOA
- Products that drive organizational structure
 - effects on services
- Cultural differences
 - Individuals
 - Corporate
 - National
- Outsourcing

Conclusion

There are many non-technical forces that can affect the technical architecture of a system. The more the focus group discussed these forces the more forces the group identified. In the face of so many powerful forces it seems technical considerations are often minor considerations.

Faced with multiplying and divergent forces software architecture is pulled in many different directions. Most likely the technical architect is also pulled in many directions.

We can suggest two conclusions:

- The role of the architect is to reconcile the competing forces – whether they are technical or socio-economic.
- When judging a system architecture technical considerations alone are not sufficient. What appears as a deficient architecture may actually have been exactly the right architecture at the time it was devised.

While these conclusions may not be particularly radical they serve to help clarify what architecture is and what the role of the architect is.

